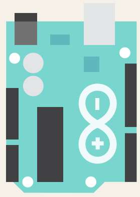
Maze Solver Project Documentation



Overview

Project Name	**MAZE SOLVER USING ARDUINO**
Project Members	Samarth Sawant (BT23ECE001) Varad Kulkarni (BT23ECE007) C Harish Raja (BT23ECE011) Vansh Patiyal (BT23ECE019) Nikhil George (BT23ECE003)
Project Dates	Start Date: Jan 22, 2025 End Date:
Background	Maze-solving robots have been a popular topic in robotics and artificial intelligence, serving as an excellent platform for learning about autonomous navigation, sensor integration, and algorithm development. These robots are designed to explore and solve a maze without human intervention, using real-time data from sensors to detect walls and paths.
Objectives	• Utilize sensors (ultrasonic) to detect walls. • Ensure the robot can solve mazes of varying complexity with minimal intervention. • Ensure smooth and precise movements using motor drivers and control mechanisms.

Introduction

Overview	**Maze-solving robots represent a significant domain in robotics, combining algorithmic decision-making with real-world navigation challenges. The concept of maze-solving dates back to ancient times, but in modern engineering, it serves as a benchmark for evaluating robotic intelligence, sensor efficiency, and algorithmic optimization. These robots autonomously navigate an unknown environment, using predefined algorithms to explore, localize, and find the shortest path to their destination.** **This project leverages Arduino-based microcontrollers to develop an efficient, autonomous maze-solving robot. The implementation includes essential algorithms such as the Wall Following algorithm and the Flood Fill algorithm, allowing the robot to traverse a maze while optimizing for the shortest path. Various hardware components, such as ultrasonic sensors, motor drivers, and infrared sensors, enable precise navigation**
Project Scope	**Autonomous Maze Navigation** – The robot will navigate through a maze using sensor feedback and decision-making algorithms. **Sensor-Based Detection** – The robot will use ultrasonic sensors to detect maze walls. **Motor Control & Movement** – The system will include motor drivers for precise control of movement and turns. **Real-Time Decision Making** – The robot will decide its path dynamically at intersections and dead ends. **Scalability** – The robot will be able to navigate mazes of varying complexity and layouts. **Arduino-Based Implementation** – The project will use an Arduino microcontroller for processing and control.
Project Constraints	**Hardware Limitations** – The project is restricted to components like Arduino, ultrasonic sensors, and motor drivers, limiting processing power and sensor accuracy. **Power Constraints** – The robot operates on battery power, which affects runtime and motor efficiency. **Maze Complexity** – The robot's ability to solve mazes is limited by sensor range, processing speed, and algorithm efficiency.

Size Constraints – The robot's dimensions must allow it to navigate

Research papers referred.

1) A Review on Maze Solving Robot Using Arduino Uno" (Shri Chhatrapati Shivaji Maharaj QIP Conference, 2019)

2) A Versatile Approach for Teaching Autonomous Robot Control" (IEEE, 2017)

3) Autonomous Maze Solving Robot Using Arduino" (IJARET, 2021)

4) An Efficient Algorithm for Robot Maze-Solving" (IEEE, 2010)

Required Materials

MATERIALS	*COST*
1. Tyres - Qty: 2	₹100
2. Rechargeable Battery - Qty: 1	₹600
3. 12V DC Motor - Qty: 2	₹440
4. Two Steel Clamps - Qty: 2	₹90
5. Adapter - Qty: 1	₹170
6. Extra Wire (Battery) - Qty: 1	₹20
7. Connecting Wires (1m) - Qty: 2	₹30
8. Ultrasonic Sensors - Qty: 3	₹220
9. LED - Qty: 3	₹21
10. Screwdriver - Qty: 1	₹20
11. Jumper Wires - Qty: 30	₹58
12. L298N Motor Driver - Qty: 1	₹105
13. Sandpaper - Qty: 2	₹40
14. Arduino Shell - Qty: 1	₹130
15. Arduino - Qty: 1	₹500
16. Wire Stripper/Cutter	₹71
TOTAL COST:	₹2615

Hardware Specifications

ARDUINO UNO R3:

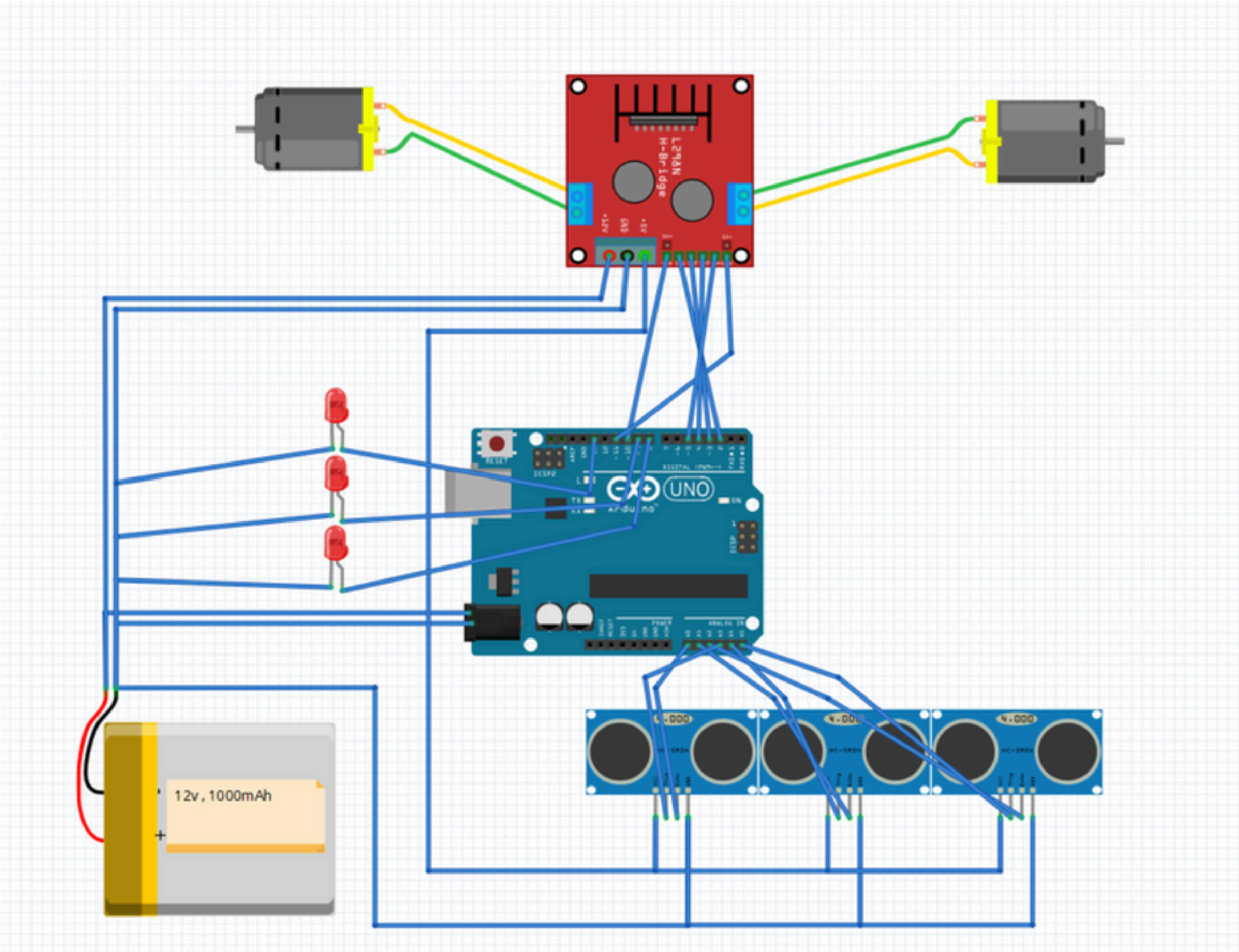
Specification	*Value*	*Notes*
Microcontroller	*ATmega328P*	*8-bit AVR*
Clock Speed	*16 MHz*	*External crystal*
Flash Memory	*32 KB*	*~0.5 KB bootloader*
SRAM	*2 KB*	*For runtime storage*
EEPROM	*1 KB*	*Non-volatile memory*
Digital I/O Pins	*14*	*6 support PWM output*
Analog Input Pins	*6*	*10-bit ADC resolution*
Operating Voltage	*5 V*	*Logic level for MCU*

Input Voltage	*7-12 V*	*6-20 V acceptable range*
USB Interface	*ATmega16U2*	*USB-to-serial conversion*
Timers	*Timer (0, 1, 2)*	*(delays, PWM, interrupts)*
Delay Function	*delay()*	*Blocking delay wTimer0*
Instruction Cycle	*~62.5 ns*	*At 16 MHz clock speed*
Programming Env.	*Arduino IDE (C/C++)*	*Wiring framework*
Communication Proto.	*UART, SPI, I2C*	*For interfacing ext devices*

L982N MOTOR DRIVER MODULE:

Specification	*Value*	*Notes*
Driver IC	*L298N*	*Dual H-bridge*
Channels	*2*	*2 DC motors*
Motor Voltage	*5-35 V*	*Motor supply range*
Logic Voltage	*5 V*	*TTL logic level*
Max Current per Channel	*~2 A*	*Up to ~3 A peak*
Total Current	*~4 A*	*Combined output*
PWM Frequency	*~20 kHz*	*For speed control*
Protection	*Flyback Diodes*	*EMF protection*
Operating Temp	*0°C to 70°C*	*Typical range*
Dimensions	*~43 x 43 mm*	*Module size*

Circuit Diagram


L298N
H-Bridge
UNO
12v , 1000mAh

Project Timeline

Task	*Date Completed*	*Notes*
Brainstorm and come up with an idea for the project which will increase our understanding of electronic devices and circuits.	Jan 22, 2025	**MAZE SOLVER USING ARDUINO**
Find all the information required to start the project, design the circuit and get ready with all the materials required.	Jan 29, 2025	Circuit Design Completed.
Learn about the electronics used in the project. Take note of all necessary precautionary measures to keep the electronics and ourselves safe.	Feb 5, 2025	specifications and cautions noted.
Starting with testing of all components, making a base structure to start off with placement of the components to maximize utilization of resources.	Feb 12, 2025	base structure built.

Conclusion

Project Outcomes	1. **Functional Maze-Solving Robot**: Designed and built a robot to navigate mazes autonomously. 2. **Arduino Expertise**: Gained hands-on experience with Arduino programming and integration. 3. **Sensor Utilization**: Learned practical use of sensors for wall detection. 4. **Team Management Skills**: Enhanced collaboration and project management abilities. 5. **Power Management**: Efficiently managed hardware power requirements. 6. **Debugging**: Improved error detection and troubleshooting skills. 7. **Documentation**: Developed professional-level project documentation. 8. **Cost-Effective Design**: Built a budget-friendly functional prototype. 9. **Future Projects**: Established a foundation for advanced robotics projects.